

EDUCATION

- **IT University of Copenhagen** Copenhagen, Denmark
Master of Science in Computer Science July 2024 – June 2026
- **University of Copenhagen** Copenhagen, Denmark
Bachelor of Science in Computer Science July 2021 – June 2024
- **University of Southern Denmark** Denmark
Elite Summer School - Robotics & Entrepreneurship (3-week intensive) Aug. 2025

EXPERIENCE

- **DISCO-2 CubeSat Project** Copenhagen, Denmark
Embedded Systems Developer Jan 2025 - Present
 - **Heterogeneous SoC Architecture:** Developed software for Cortex-A53/Cortex-M7 heterogeneous system; A53 runs Linux for image processing while M7 handles low-power operations during system suspend.
 - **Yocto BSP Development:** Built custom embedded Linux distribution using Yocto/OpenEmbedded for 3U CubeSat; configured kernel, device trees, and minimal rootfs for resource-constrained space environment.
 - **Camera Pipeline:** Implemented camera control application interfacing with onboard cameras via CSI; developed configurable image processing pipeline (DIPP) with multiple processing stages for in-orbit image analysis.
 - **CSP Communication:** Integrated CubeSat Space Protocol (CSP) stack for telemetry and image downlink to ground station; implemented packet fragmentation and error recovery for reliable transmission.
 - **Cross-Platform Debugging:** Debugged image pipeline issues using GDB remote debugging on target hardware; traced memory and timing issues across A53/M7 boundary.
- **PeopleNet A/S** Copenhagen, Denmark
Software Developer (Part-time) Aug 2022 - Present
 - **Enterprise Development:** Built SharePoint Framework (SPFx) solutions with Microsoft Graph API and PnP JS; developed Azure Functions for backend processing and automation.

PROJECTS

- **LEGO Quality Control System (Collaboration with LEGO):** Computer vision system for automated scratch detection using AnomalyDINO. Developed during Elite Robotics Summer School working with collaborative industrial robots.
- **Mesh Grid Networks (BSc Thesis):** Distributed mesh networking protocol implementation; analyzed routing algorithms and packet delivery in decentralized communication systems.

PUBLICATIONS

- **COPA2024, San Diego:** “Building Towards a Distributed, Dynamic Solution to the Santa Claus Problem” - Co-authored with David Marchant. Concurrent distributed systems with performance analysis.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Bash
- **Embedded Platforms:** ARM Cortex-A53, ARM Cortex-M7, Yocto/OpenEmbedded, Linux kernel, device trees
- **Protocols & Interfaces:** CSP (CubeSat Space Protocol), CSI/MIPI, SPI, I2C, UART, TCP/IP
- **Tools:** GCC cross-compilation, GDB remote debugging, Make/CMake, Git, Docker

LANGUAGES

- **Danish:** Native
- **English:** Fluent
- **Arabic:** Fluent